Q-Bench-Video: Benchmark the Video Quality Understanding of LMMs

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https://github.com/Q-Future/Q-Bench-Video

Abstract

With the rising interest in research on Large Multi-modal Models (LMMs) for video understanding, many studies have emphasized general video comprehension capabilities, neglecting the ***systematic exploration into video quality understanding****. To address this oversight, we introduce* ***Q-Bench-Video*** *in this paper, a new benchmark specifically designed to evaluate LMMs' proficiency in discerning video quality.* ***a)*** *To ensure video source diversity,* ***Q-Bench-Video*** *encompasses videos from natural scenes, AI-generated content (AIGC), and computer graphics (CG).* ***b)*** *Building on the traditional multiple-choice questions format with the Yes-or-No and What-How categories, we include Open-ended questions to better evaluate complex scenarios. Additionally, we incorporate the* ***video pair quality comparison*** *question to enhance comprehensiveness.* ***c)*** *Beyond the traditional Technical, Aesthetic, and Temporal distortions, we have expanded our evaluation aspects to include the dimension of AIGC distortions, which addresses the increasing demand for video generation. Finally, we collect a total of 2,378 question-answer pairs and test them on 12 open-source & 5 proprietary LMMs. Our findings indicate that while LMMs have a foundational understanding of perceptual video quality, their performance remains incomplete and imprecise, with a notable discrepancy compared to the performance of human beings. Through* ***Q-Bench-Video****, we seek to catalyze community interest, stimulate further research, and unlock the untapped potential of LMMs to close the gap in video quality understanding.*

1. Introduction

As the field of artificial intelligence (AI) continues to evolve, Large Multi-modal Models (LMMs) [4, 16, 19, 50, 51] are progressively utilized in high-level video understanding tasks. These models have shown remarkable capabilities in analyzing and interpreting the semantic content of videos, such as classifying objects, identifying actions, and recognizing events. However, **the aspect of video quality, which is vital for optimizing compression and transmission systems, enhancing viewer experience, and establishing standards for high-quality video generation, has received less attention.** Although numerous LMM video benchmarks[8, 10, 23] have been developed to assess the semantic understanding of videos by LMMs comprehensively, benchmarks systematically targeting video quality are still lacking. Additionally, while semantic understanding is closely linked to high-level video information, the perception and understanding of low-level information are crucial in video quality [5, 25] as well.

To address this gap, we introduce **Q-Bench-Video**, a novel benchmark specifically designed to systematically evaluate the video quality understanding of LMMs. As illustrated in Fig. 1, our benchmark encompasses a wide range of video content, including natural scenes, AI-generated content (AIGC), and computer graphics (CG), ensuring diversity in video sources. In addition, to maintain a reasonable distribution of source video quality, we employ uniform sampling from video datasets that contain subjective quality annotations. This approach guarantees comprehensive coverage of the quality spectrum while avoiding imbalanced quality distributions. Moreover, we extend beyond traditional video evaluations by incorporating both multiple-choice questions (MCQs) and open-ended questions. This enables a more thorough analysis of LMMs' ability to discern video quality across diverse scenarios. We further introduce a new evaluation dimension specifically tailored to assess distortions related to AIGC, which are increasingly prominent in video generation tasks. Recognizing the importance of quality comparison settings in real-world applications, such as camera parameter optimization and AIGC video generation, we further incorporate video pairs to facilitate quality comparison assessment as well. In total, we collect 1800 videos and annotated 2,378 question-answer pairs for validation, creating a robust framework for systematic evaluation on video quality understanding.

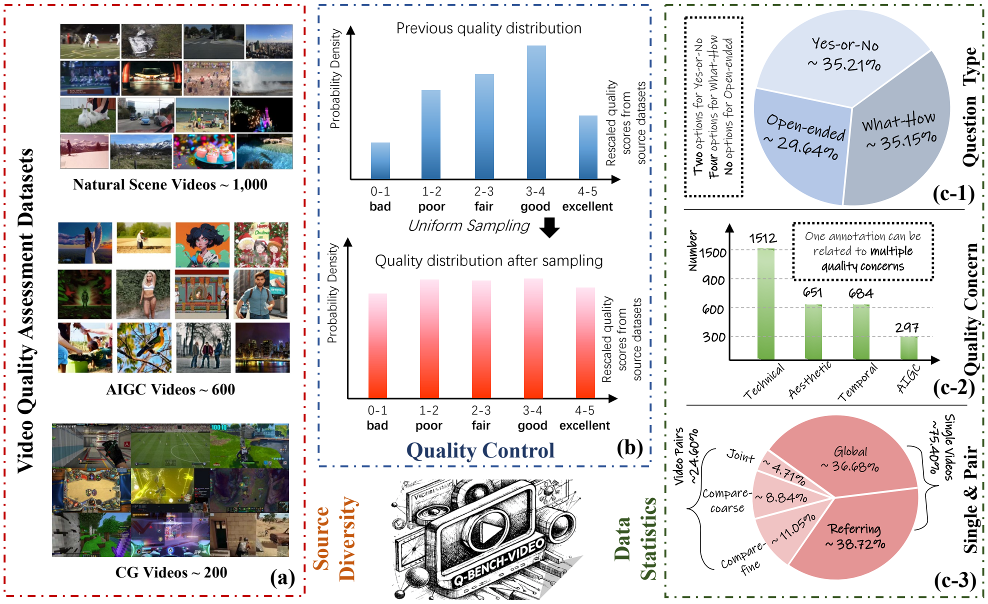


Figure 1. The construction overview of the proposed **Q-Bench-Video**. To ensure **diversity in video content**, we collect natural scenes, AIGC, and CG videos from video quality assessment datasets as depicted in (a). To achieve a balanced quality distribution among the sampled videos, we employ **uniform sampling for quality control**. As indicated in (c-1) and (c-2), we utilize three types of questions (*Yes-or-No, What-How, Open-ended*) and address a comprehensive range of quality concerns including *Technical*, *Aesthetic*, *Temporal*, and *AIGC* distortions. Additionally, we incorporate the video pairs comparison task to enhance the comprehensiveness of the benchmark.

Through the rigorous experiment, we demonstrate that while LMMs show promise in video quality assessment, their performance lags significantly behind human-level understanding. By offering a systematic and thorough evaluation of LMMs' video quality perception, **Q-Bench-Video** aims to foster research in this underexplored area and push the boundaries of LMM capabilities on video quality understanding. Our contributions can be summarized as follows:

- We introduce **Q-Bench-Video**, the first comprehensive benchmark explicitly designed to assess the video quality understanding capabilities of LMMs. This benchmark includes a diverse collection of source videos and ensures a balanced quality distribution, complemented by human-crafted question-answer annotations.
- Our evaluation framework spans four key quality dimensions: *Technical*, *Aesthetic*, *Temporal*, and *AIGC* distortions, which offers a holistic evaluation approach to video quality assessment. Uniquely, **Q-Bench-Video** enhances its utility by introducing the task of **video pairs comparison**, which sets it apart from existing video benchmarks.
- We conduct a comprehensive evaluation using both proprietary and open-source LMMs to measure their effectiveness in understanding video quality. The results expose notable deficiencies in current LMMs, while also shedding light on performance variations across different quality dimensions. These findings provide critical insights and suggest promising directions for future enhancements in the field of video quality understanding.

2. Related Works

2.1. Video LMMs and Benchmarks

The rapid advancement of Large Multi-modal Models (LMMs) in recent years [4, 28–30, 52, 56, 67] has showcased their remarkable perception and cognitive abilities across various multimodal benchmarks for images [9, 33, 35, 47, 62]. As development progresses, the focus of visual analysis has gradually shifted from images to videos. Early efforts [22, 27, 31, 50] aimed at unlocking the video understanding potential of LMMs have yielded promising results. However, initial video-based benchmarks [24, 34, 41] typically concentrated on specific aspects of video comprehension, falling short of fully capturing the performance of these models due to limitations such as a lack of diversity in video types and inadequate coverage of temporal dynamics. In response, recent video benchmarks [8, 10] have moved toward a more comprehensive evaluation of LMMs. Nonetheless, these efforts focus on high-level semantic understanding without systematic exploration of video quality.

2.2. Video Quality Assessment

Video Quality Assessment (VQA) is a task aimed at quantifying video scores based on visual quality. Initially, early VQA methods employ hand-crafted features extracted from videos and regress the features into quality scores [17, 20, 26, 40, 58, 64, 65]. With the emergence of deep neural networks, a shift occurs as numerous methods adopted deep

Table 1. Overview of the diverse video source datasets in the **Q-Bench-Video**. We consider various video content types, including **natural scenes, AIGC, and CG videos**. The term 'MOS' denotes that the videos are annotated via Mean Opinion Scores under ITU standards [1]. We have conducted uniform sampling based on their quality labels to ensure a **balanced quality distribution**.

Video Type	Video Source Dataset	MOS	Quality Concerns	Sampled Size	Full Dataset Size
Natural (1000)	LSVQ [54]	✓	Spatial & Temporal	600	39K
	MaxWell [46]	✓	Spatial & Temporal & Aesthetic	350	4.5K
	WaterlooSQoE-III [6]	✓	Quality-of-Experience	20	450
	WaterlooSQoE-IV [7]	✓	Quality-of-Experience	30	1,350
AIGC (600)	T2VQA-DB [17]	✓	Quality & Text Alignment	200	10K
	VideoFeedback [13]	×	Quality & Text Alignment	400	37.6K
CG (200)	LIVE-YT-Gaming [55]	✓	Visual Quality	200	600

learning techniques for VQA tasks [18, 21, 38, 42]. As the field progresses, newer methods begin incorporating considerations for both temporal dynamics and aesthetic qualities, leading to a more holistic approach to video quality analysis [3, 12, 43, 45, 59]. Moreover, the evolution of large-scale models has further revolutionized VQA methodologies. Many recent approaches have redefined the traditional quality assessment process into a quality question-answering format [11, 15, 48, 60, 63]. This adaptation leverages the substantial prior knowledge embedded in large models to enhance the precision of quality quantification [61]. Despite these technological advances, VQA still grapples with challenges in providing interpretable quality scores and deepening the understanding of how models perceive and analyze video quality more comprehensively.

3. Benchmark Construction

3.1. Benchmark Principle

The **Q-Bench-Video** is designed based on three guiding principles: (**1**) **It encompasses a broad spectrum of video content**, including natural scenes, AIGC, and CG videos. (**2**) **It ensures a comprehensive and representative sampling process across a wide quality range**, enhancing the benchmark's overall effectiveness. (**3**) **It primarily focuses on the aspects of video quality that significantly influence the viewing experience**, including technical, aesthetic, temporal, and AIGC distortions. This significantly differs from other video benchmarks that prioritize semantic understanding. Additionally, the **video pair quality comparison** is integrated to address the challenges associated with comparing video quality. The overall construction process can be clearly overviewed in Fig. 1.

3.2. Source Videos Collection

As shown in Table 1, the source videos are primarily gathered from video quality assessment datasets. We selected videos from these datasets for two main reasons: (**1**) These datasets have inherently considered the diversity of quality features during video selection; (**2**) These datasets possess quality annotations that adhere to ITU standards [1] (with the exception of VideoFeedback), which allow us to accurately and authentically sample videos.

Our sampling method primarily employs a uniform approach, extracting videos evenly from each dataset based on the quality range. Moreover, considering the current popularity of AIGC and CG videos, we have also incorporated a selection of these video types. (*More details in the Supp.*)

3.3. Benchmark Designs

In this section, we provide a detailed description of the design of **Q-Bench-Video**. In this benchmark, the meta-structure tuple (`V,Q,A,C`) of each data item can be decomposed into several components: the video object `V` (which can be a single video or a pair of videos), the video quality query `Q`, the set of possible answers `A`, and the correct answer `C`. The question samples are listed in Fig. 2.

3.3.1. Question Types

I) Yes-or-No Questions. The basic *Yes-or-No* questions are designed to prompt LMMs to make binary judgments on video quality queries, typically limited to the answers *Yes* or *No*. To address the potential bias in LMMs that may skew towards *yes* responses, we employ an annotation adjustment process, which modifies the questions and correct answers if necessary (based on the *Yes-or-No* ratio). This process ensures that the distribution of correct answers, either *Yes* or *No*, remains balanced at about 50%:50% ratio in the end. This balanced approach allows for a more accurate assessment of LMMs' performance on *Yes-or-No* questions.
II) What-How Questions. The *What-How* questions are commonly utilized in benchmarks for LMMs. The *What* questions focus on identifying specific distortions (*e.g., What is the most apparent distortion in this video?*). On the other hand, the *How* questions are employed to distinguish the finer details of distortion levels (*e.g., How is the overall clarity of this video?*). Including both *What* and *How* questions allows **Q-Bench-Video** to thoroughly and meticulously evaluate LMMs' ability on identifying video distortions and evaluating the distortion levels.
III) Open-ended Questions. It's important to note that the two types of questions previously mentioned require LMMs to select the correct answer from a predefined set. However, in many real-world scenarios, *Open-ended Questions*, which do not restrict responses to a predefined set, are often more necessary and challenging for LMMs (*e.g., What*

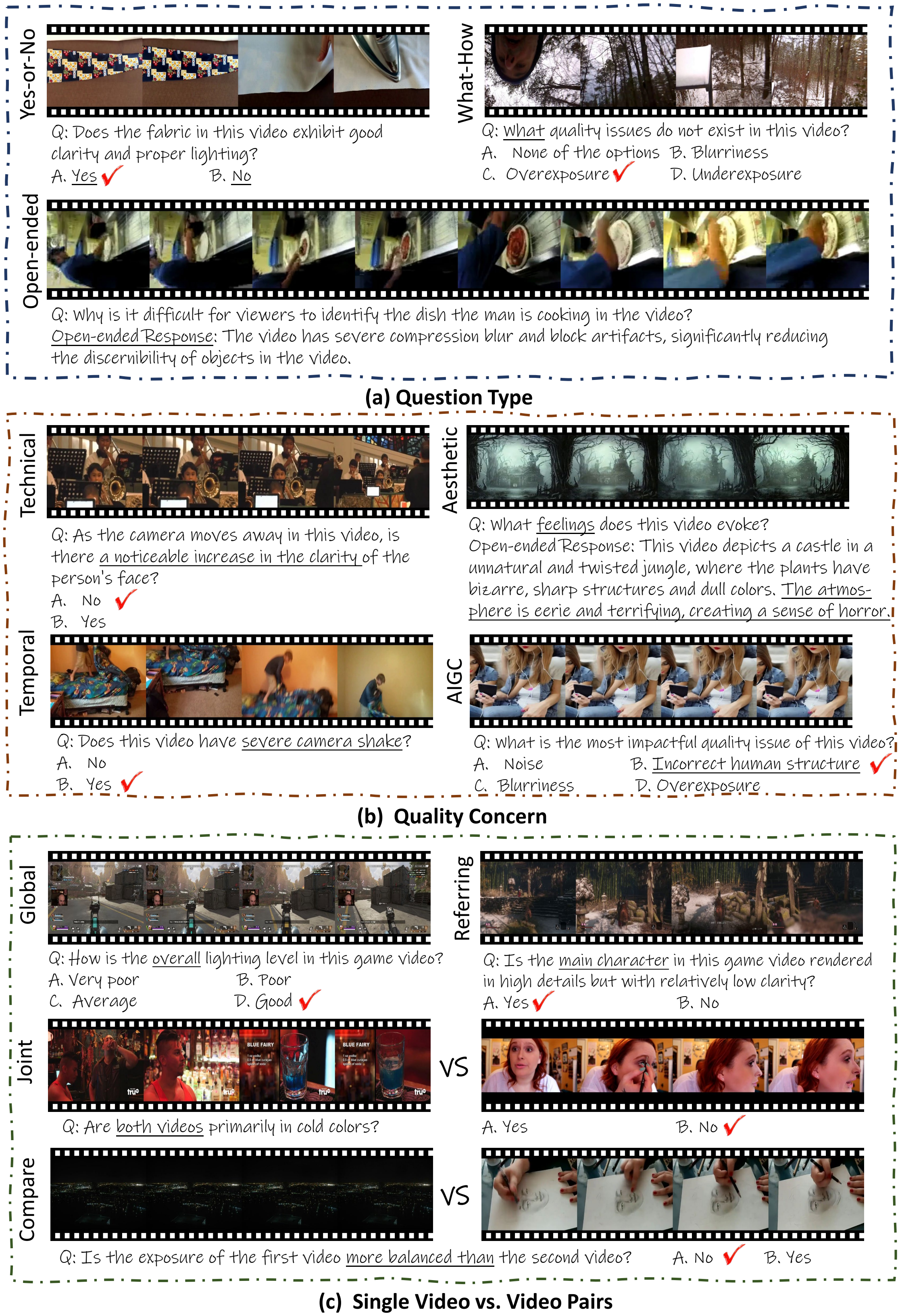


Figure 2. The visualization samples from **Q-Bench-Video**, with the question-answer content most representative of each subcategory being underlined. It is important to note that, regarding quality concerns, a single question-answer annotation may not only focus on one distortion dimension. Therefore, the distortion visualization examples shown in (b) primarily highlight instances that are most closely aligned with the mentioned distortion types.

are the possible factors that lead to the low clarity of this video? Please list and explain.). By adopting this form of questioning, we can better assess an LMM's ability to perceive video quality in real-world conditions.

3.3.2. Quality Concerns

It's important to recognize that video quality can be influenced by multiple factors on some occasions. Therefore, a query tuple (`V,Q,A,C`) **does not need to be restricted to a single concern. It can address multiple concerns simultaneously.** For instance, the question *Is this video clear and well-composed?* can be seen as evaluating both technical and aesthetic quality understanding.

I) Technical Distortions. Technical distortions refer to the *low-level degradation* in video quality that arises from the limitations of recording, compression, and transmission [37, 54]. These distortions often include artifacts such as *blurring, noise, compression artifact, exposure, etc.*, which are directly tied to the technical processes used in video production and delivery. The CG-specific distortions related to geometry and texture are taken into consideration.

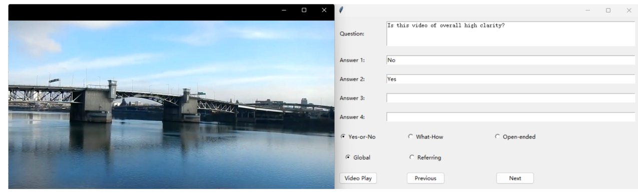
II) Aesthetic Distortions. Aesthetic distortions involve deviations from the *intended visual style, artistic design, or creative intent that negatively affect the viewer's perception* of the video [14, 44]. These distortions can include aspects such as *confusing color, poor composition, lighting inconsistencies, or distracting elements* that reduce the overall aesthetic appeal. Unlike technical distortions, aesthetic distortions are subjective and might be affected by viewer preference, cultural context, or artistic norms.

III) Temporal Distortions. Temporal distortions are related to the *degradation of visual quality over time*, impacting the fluidity and consistency of the video [36]. These distortions manifest as issues like *screen shake, flickering, motion inconsistency, frame drops, and stuttering* that result from unstable shooting devices, dramatically changing lighting conditions, and unstable bitrate environments [46]. Such disruptions hinder the viewer's natural perception of the video, leading to a disjointed and unpleasant viewing experience, especially in live-streaming scenarios.

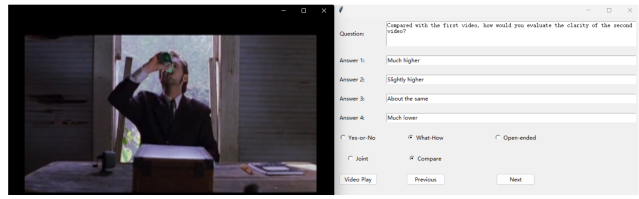
IV) AIGC Distortions. AIGC distortions pertain to *imperfections and unnaturalness specifically arising from the generation of video content through AI models* [32, 61]. These distortions may include *unnatural textures, inconsistent lighting, uncanny facial features, or unrealistic object behavior* that result from limitations or biases in the training data, model architecture, or generative process. These distortions are unique to AI-generated content and require specialized evaluation metrics that consider both the technical and perceptual quality aspects.

3.3.3. Single Videos & Video Pairs

Accurately comparing and jointly analyzing the quality of video pairs is sometimes more crucial than assessing the quality of a single video, especially in scenarios such as performance tests in video compression and quality control in video generation (In which it is more important to find out *Which video is better in visual quality and why?*) [49, 66]. Therefore, in **Q-Bench-Video**, we include video quality queries for both **single videos** and **video pairs**.



(a) Annotation GUI for single videos of **Q-Bench-Video**.

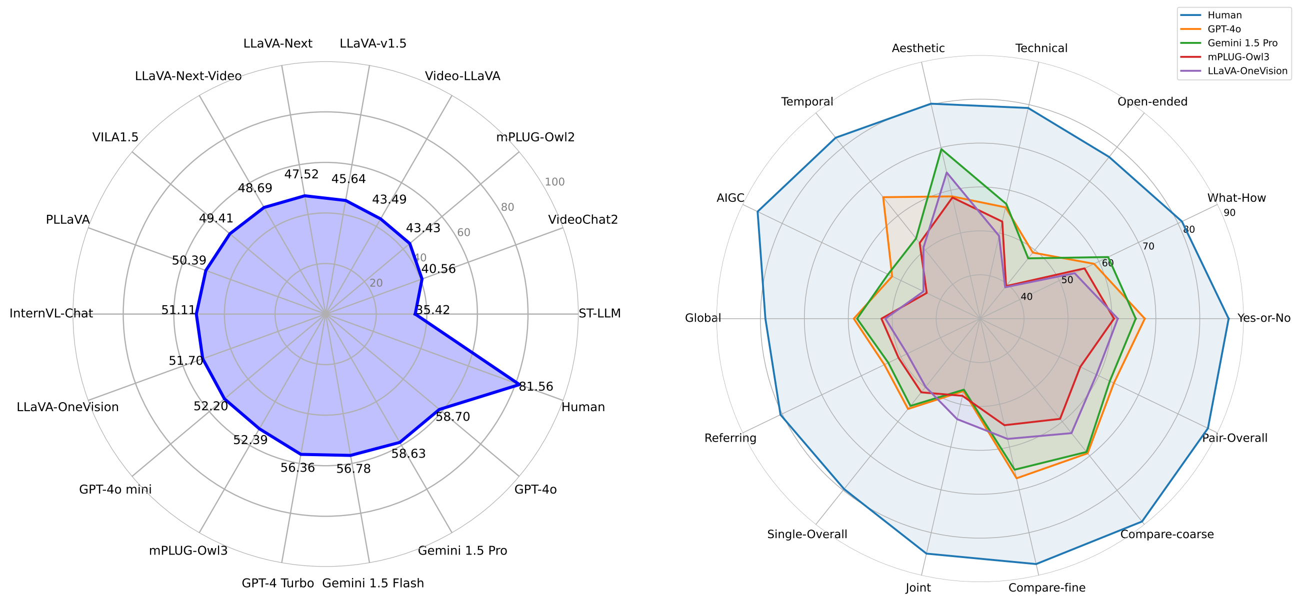


(b) Annotation GUI for video pairs of **Q-Bench-Video**.

Figure 3. Illustration of the annotation GUIs for **Q-Bench-Video**. (a) shows the interface for annotating single videos, where the annotator can select the question type and play the videos using the *Video Play* button. The annotator can also switch to the next and previous annotation with the *Next* and *Previous* buttons. (b) presents the interface for annotating video pairs. When the annotator presses the *Video Play* button, the video pairs are played sequentially, with a five-second gray screen serving as an interval between the two videos.

I) Single Videos. Queries related to single videos can primarily be categorized into two types: **a**) **Global perception**, which involves questions about the overall visual quality of the video, such as *How is the overall contrast of this video?* **b**) **Referring perception**, which focuses on the visual quality of specific elements within the video, like querying *What is the most apparent distortion when the player strikes the ball?* Through these approaches, we aim to comprehensively evaluate the LMMs' ability to perceive both the overall and localized aspects of video quality.

II) Video Pairs. Firstly, to ensure that the comparison of video pairs is clear and meaningful, **comparisons are only made between videos from the same source**, such as videos from natural sources being paired together while CG videos and AIGC videos are not being paired. There are mainly two types of video pair categories: **a**) **Joint analysis**, which involves understanding the shared quality features of the video pairs, for example, such as asking *Are both videos blurry?* **b**) **Comparative analysis**, which involves comparing the quality dimension across two videos, such as *How does the level of brightness in the first video compare to that in the second one?* It is important to note



(a) Overall performance on **Q-Bench-Video**.

(b) Subcategory performance on **Q-Bench-Video**.

Figure 4. A concise summary of the LMMs' performance on **Q-Bench-Video**. (a) provides a comparison detailing the overall performance of humans and 17 selected LMMs, including both *proprietary* and *open-source* models. (b) illustrates a radar chart that outlines the performance of the **top-2** *proprietary* LMMs (GPT-4o & Gemini 1.5 Pro) and *open-source* LMMs (mPLUG-Owl3 & LLaVA-OneVision) across various subcategories within **Q-Bench-Video**.

that we further categorize comparisons based on the difference in quality labels of the videos involved into **coarse-grain** (with relatively more significant visual quality differences) and **fine-grain** (with relatively minor visual quality differences) comparisons. (*More details in the Supp.*)

3.3.4. Questions & Answers Annotation

The annotation process of **Q-Bench-Video** is conducted in a well-controlled laboratory environment. A total of 8 experts are employed and trained to ensure the consistency of the annotations. The experts are required to watch the videos in their entirety before making annotations. Each annotated question-answer pair is then reviewed by at least three other experts to ensure its validity and accuracy.

3.4. Benchmark Setting & Evaluation

Unless specifically stated otherwise, for *Video LMMs* we typically analyze by uniformly sampling 16 frames from the video, while for *Image LMMs*, the sampling is reduced to 8 frames. It's worth noting that if the LMMs are not compatible with the specified number of sampling frames (e.g. if they cannot support up to 8-frame input), we will use the official default settings for testing. For **Yes-or-No** and **What-How** questions, if the LMMs can accurately respond with the options, we directly record the accuracy of the responses as results. If the LMMs cannot provide option-based answers, we implement a GPT-assisted evaluation strategy to help judge the accuracy of the answers. For **Open-ended** questions, since the answers are open-ended and cannot be directly quantified for accuracy, we also employ the GPT-assisted evaluation strategy. This involves GPT scoring the responses based on their accuracy, completeness, and relevance compared to the annotated answer.

4. Results of Q-Bench-Video

4.1. Experimental Setting

LMMs Participants. A total of 17 LMMs (12 *Open-source LMMs* and 5 *Proprietary LMMs*) are included for validation, which includes **a**) 3 *Open-source Image LMMs*: LLaVA-Next [30], LLaVA-v1.5 [28], and mPLUG-Owl2 [53]; **b**) 9 *Open-source Video LMMs*: mPLUG-Owl3 [51], LLaVA-OneVision [19], InternVL-Chat [4], VILA1.5 [16], PLLaVA [50], LLaVA-Next-Video [57], ST-LLM [31], Video-LLaVA [27], and VideoChat2 [22]; **c**) 5 *Proprietary LMMs*: Gemini 1.5 Flash, Gemini 1.5 Pro [39], GPT-4o mini, GPT-4o, and GPT-4 Turbo [2].

Subsets Split. The **Q-Bench-Video** is divided into `test` (1,186 question-answer items) and `dev` (1,192 question-answer items) subsets. The correct answers will be released and proprietary for the `dev` and `test` subsets. All discussions and analyses are based on the `test` subset.

Table 2. Results on the `test` subset for the video quality perception ability of LMMs. The best performance is marked in **bold** and the second performance is underlined for *Open-source* and *Proprietary* LMMs respectively. The *Open-ended* questions are excluded for **Random guess**. (*Performance of the* `test` *subset is in the Supp.*)

Sub-categories	**Question Types**			**Quality Concerns**				
LMM *(LLM)*	*Yes-or-No*↑	*What-How*↑	*Open-ended*↑	*Tech.*↑	*Aes.*↑	*Temp.*↑	*AIGC*↑	*Overall*↑
Random guess *w/o Open-ended*	50.00%	25.00%	/	37.10%	37.31%	37.25%	37.22%	37.79%
Human	86.57%	81.00%	77.11%	79.22%	80.23%	82.72%	86.21%	81.56%
Open-source Image LMMs								
LLaVA-Next (*Mistral-7B*)	62.83%	45.14%	33.69%	46.38%	57.86%	47.84%	48.46%	47.52%
LLaVA-v1.5 (*Vicuna-v1.5-13B*)	52.98%	46.44%	37.01%	45.77%	58.12%	45.30%	46.48%	45.64%
mPLUG-Owl2 (*LLaMA2-7B*)	59.19%	39.07%	31.19%	42.07%	52.38%	41.71%	39.37%	43.43%
Open-source Video LMMs								
mPLUG-Owl3 (*Qwen2-7B*)	60.48%	**56.39%**	39.48%	**52.68%**	58.31%	**52.05%**	43.49%	**52.39%**
LLaVA-OneVision (*Qwen2-7B*)	61.34%	53.88%	39.15%	49.35%	**64.15%**	50.68%	44.30%	51.70%
InternVL-Chat (*Vicuna-7B*)	**66.02%**	52.13%	33.93%	48.42%	52.73%	50.59%	53.12%	51.11%
VILA1.5 (*LLaMA3-8B*)	61.95%	46.00%	**39.60%**	47.85%	57.85%	45.65%	42.57%	49.41%
PLLaVA (*Mistral-7B*)	65.63%	52.33%	32.23%	49.69%	61.32%	50.96%	**53.64%**	50.39%
LLaVA-Next-Video (*Mistral-7B*)	61.34%	45.95%	38.10%	49.03%	60.94%	46.97%	49.40%	48.69%
ST-LLM (*Vicuna-v1.1-7B*)	44.63%	28.50%	32.78%	34.99%	46.11%	34.28%	34.02%	35.42%
Video-LLaVA (*Vicuna-v1.5-7B*)	64.67%	40.79%	29.11%	43.25%	54.04%	42.38%	42.76%	43.49%
VideoChat2 (*Mistral-7B*)	56.09%	29.98%	34.99%	39.26%	50.02%	38.25%	35.88%	40.56%
Proprietary LMMs								
Gemini 1.5 Flash	65.48%	56.79%	47.51%	54.11%	66.58%	53.51%	50.22%	56.78%
Gemini 1.5 Pro	65.42%	**62.35%**	47.57%	**56.80%**	**69.61%**	53.38%	**53.26%**	58.63%
GPT-4o mini	62.95%	50.93%	42.10%	49.38%	60.90%	48.43%	41.71%	52.20%
GPT-4o	**67.48%**	58.79%	**49.25%**	56.01%	58.57%	**65.39%**	52.22%	**58.70%**
GPT-4 Turbo	66.93%	58.33%	40.15%	54.23%	66.23%	54.00%	52.04%	56.36%

4.2. Findings

The overall performance and subcategory comparisons (human vs. top-performing LMMs) on **Q-Bench-Video** can be quickly glanced at Fig. 4. Detailed performance across the subcategories for each LMM are shown in Table 2 (**Question Types** & **Quality Concerns**) and Table 3 (**Single Videos vs. Video Pairs**) respectively. The findings are organized as follows:

1) General Performance. Human>Proprietary LMMs>Open-source LMMs>Random guess. From the performance results presented in Table 2, we observe that nearly all LMMs significantly outperform *random guess*, demonstrating their basic capability to understand video quality. Among the open-source LMMs, the recently released mPLUG-Owl3 achieves the highest *overall* performance at 52.39%, even slightly surpassing GPT-4o mini (52.20%), followed closely by LLaVA-OneVision (51.70%) and InternVL-Chat (51.11%). Image LMMs deliver moderate performance. Although they outperform some Video LMMs, the gap between them and the latest Video LMMs is still notable. Benefiting from larger training datasets and more parameters, proprietary LMMs (except GPT-4o mini) outperform all open-source models. However, even the best-performing model, GPT-4o, which achieved an *overall* performance of 58.70%, still lags behind human performance by 22.86%. This gap highlights that, despite the advancements in current state-of-the-art LMMs on video general understanding, there remains a significant need for improvement in video quality understanding ability.

2) Question Types. *Open-ended* questions are more challenging for LMMs. From Table 2, a discernible hierarchy in task difficulty for video quality assessment emerges for both humans and LMMs, arranged as follows: *Open-ended* >*What-How* >*Yes-or-No*. It is crucial to highlight that while humans exhibit a performance decline in *Open-ended* tasks by approximately 9.46% compared to *Yes-or-No* tasks, and about 3.89% compared to *What-How* tasks, these reductions are markedly less pronounced than those observed in LMMs for the *Open-ended* questions. This disparity underscores a significant proficiency gap between LMMs' capability in handling straightforward, closed-form questions and their effectiveness in navigating the complexities of real-world problem-solving on open-ended questions, particularly in the context of video quality evaluation.

3) Quality Concerns. LMMs exhibit unbalanced performance across different types of distortions. From Table 2, it is evident that humans are particularly good at identifying *AIGC* distortions, while LMMs demonstrate stronger performance in detecting *Aesthetic* distortions. This distinction likely stems from the inherent sensitivity of humans to the conspicuous unnaturalness of *AIGC* distortions, which readily draws human attention. In contrast, *Aesthetic* distortions, which often involve high-level semantic nuances, align more closely with the training contexts of LMMs, enabling them to excel in this area. However, LMMs face challenges with *AIGC* distortions due to insufficient exposure to such anomalies during their pretraining phases, specific architectural constraints, and imperfections

Table 3. Results on the `test` subset for the video quality perception ability across single videos and video pairs of LMMs. The best performance is marked in **bold** and the second performance is underlined for *Open-source* and *Proprietary* LMMs respectively.

Sub-categories	**Single Videos**			**Video Pairs**			
LMM *(LLM)*	*Global*↑	*Referring*↑	*Overall*↑	*Joint*↑	*Compare -fine*↑	*Compare -coarse*↑	*Overall*↑
Random guess *w/o Open-ended*	36.49%	38.61%	37.71%	40.56%	37.94%	36.83%	38.00%
Human	78.87%	80.43%	79.65%	84.90%	87.34%	89.11%	87.56%
LLaVA-Next (*Mistral-7B*)	51.33%	50.20%	50.73%	38.03%	48.00%	42.48%	43.46%
LLaVA-v1.5 (*Vicuna-v1.5-13B*)	47.99%	51.94%	50.10%	27.72%	34.60%	42.12%	36.42%
mPLUG-Owl2 (*LLaMA2-7B*)	46.86%	43.51%	45.07%	51.49%	37.10%	40.28%	43.69%
Open-source Video LMMs							
mPLUG-Owl3 (*Qwen2-7B*)	**52.46%**	50.60%	51.47%	48.03%	54.90%	59.20%	55.31%
LLaVA-OneVision (*Qwen2-7B*)	51.56%	48.43%	49.89%	53.48%	**58.10%**	**63.36%**	**59.41%**
InternVL-Chat (*Vicuna-7B*)	51.15%	51.86%	51.52%	48.85%	51.10%	49.20%	49.79%
VILA1.5 (*LLaMA3-8B*)	52.35%	47.37%	49.69%	56.11%	45.40%	48.04%	48.84%
PLLaVA (*Mistral-7B*)	51.44%	**55.49%**	**53.60%**	40.36%	50.40%	54.16%	49.90%
LLaVA-Next-Video (*Mistral-7B*)	51.33%	50.20%	50.73%	38.03%	48.00%	42.48%	43.46%
ST-LLM (*Vicuna-v1.1-7B*)	36.54%	36.49%	36.51%	28.03%	36.80%	32.08%	32.87%
Video-LLaVA (*Vicuna-v1.5-7B*)	45.46%	44.67%	45.04%	49.36%	42.00%	43.00%	44.01%
VideoChat2 (*Mistral-7B*)	43.52%	38.27%	40.72%	**57.23%**	44.40%	41.64%	45.93%
Proprietary LMMs							
Gemini 1.5 Flash	58.00%	53.18%	55.43%	46.59%	65.30%	68.84%	62.77%
Gemini 1.5 Pro	52.36%	**61.41%**	**57.19%**	45.43%	65.30%	**72.00%**	63.55%
GPT-4o mini	52.67%	48.96%	50.69%	44.00%	60.50%	63.88%	58.02%
GPT-4o	**58.75%**	54.18%	56.31%	**46.93%**	**67.30%**	69.24%	**63.80%**
GPT-4 Turbo	57.36%	52.80%	54.93%	46.13%	62.50%	64.80%	59.84%

in the generation process. In the case of proprietary LMMs, their performance on *Technical* and *Temporal* distortions appear comparably consistent, indicating a uniform capability in recognizing these two types of distortions. Nonetheless, across all four subcategories, LMMs exhibit a notable performance disparity compared to humans, with varying degrees of accuracy among the different types of distortions. This variability highlights the need for significant enhancements in LMMs' abilities to accurately understand and interpret various distortion types.

4) Single Videos vs. Video Pairs. LMMs demonstrate superior capabilities in comparing video quality. From Table 3, we observe that for single videos, LMMs achieve similar performance in *Global* and *Referring* quality perception (except for Gemini 1.5 Pro), without any significant trend of the performance for one subcategory over the other. This suggests that LMMs have comparable abilities in perceiving both *Global* video quality and *Referring* video quality. In terms of comparison, however, LMMs clearly outperform their performance on single video analysis and joint analysis. Notably, LMMs perform significantly better in the *Compare-coarse* subcategory, where video pairs have more pronounced quality differences, than in the *Compare-fine* subcategory. This highlights that LMMs are more adept at comparing video quality than analyzing the quality of single videos. This advantage in comparative assessment can be attributed to the inherent clarity in pairwise comparisons, which provide explicit contrasts, as opposed to the more ambiguous nature of evaluating a single video. Both humans and LMMs exhibit enhanced performance in comparative tasks. Although there is still a significant accuracy gap between LMMs and humans, LMMs show promising potential as effective tools for comparing video quality.

5. Limitations

1) Subjectivity can not be fully avoided in annotations for aspects like aesthetics. 2) As video generation advances, new models may show different distortions, which might potentially make parts of **Q-Bench-Video** outdated.

6. Conclusion

In this paper, we introduce **Q-Bench-Video**, the first comprehensive benchmark explicitly designed to evaluate Large Multi-modal Models' (LMMs) understanding of video quality. Our benchmark includes a diverse range of video types, questions that challenge multiple aspects of video quality, and a holistic evaluation framework encompassing *Technical*, *Aesthetic*, *Temporal*, and *AIGC* distortions. Through extensive experimentation with 17 *open-source* and *proprietary* LMMs, we find that while LMMs show promise in discerning video quality, their performance remains significantly below human-level understanding, especially when addressing *Open-ended* questions and *AIGC-specific* distortions. These findings highlight the current limitations of LMMs in video quality perception and underscore the need for further advancements in this area. By offering **Q-Bench-Video**, we aim to stimulate future research and drive improvements in the field, bridging the gap between LMM and human video quality understanding.

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